

# A further conjecture on OEIS A220031, settled by the proof already in hand

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## Abstract

OEIS A220031 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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## 1 The premise

The entry's first terms are 7, 18, 55, 137, 302, 613, 1165, 2094, ..., with offset 1. The sequence satisfies

$$a(n) = 7a(n-1) - 21a(n-2) + 35a(n-3) - 35a(n-4) + 21a(n-5) - 7a(n-6) + 1a(n-7),$$

a linear recurrence of order 7 with characteristic polynomial

$$q(x) = x^7 - 7x^6 + 21x^5 - 35x^4 + 35x^3 - 21x^2 + 7x - 1.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least 7 + 2 further terms after that index.

## 2 The further conjecture

Separately from the conjecture already settled, the entry records

$$\text{Empirical: } a(n) = (1/720)n^6 + (7/240)n^5 + (11/144)n^4 + (25/48)n^3 + (263/90)n^2 + (29/20)n - 4 \text{ for } n > 2.$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that  $a$  and the closed form  $f$  satisfy the same linear recurrence; it does not say they are the same solution of it. Write  $p$  for the annihilator of  $f$ ,

$$p(x) = x^7 - 7x^6 + 21x^5 - 35x^4 + 35x^3 - 21x^2 + 7x - 1,$$

of order  $D = 7$ . Since  $q \mid p$  and  $a$  satisfies the recurrence of  $q$ ,  $a$  satisfies that of  $p$  as well;  $f$  satisfies it by construction. Two solutions of a linear recurrence of order  $D$  with nonzero leading coefficient agree everywhere as soon as they agree at  $D$  consecutive indices, and  $a$  and  $f$  were checked to agree at  $n = 10, \dots, 16$ , which lies inside the terms the entry publishes and past the index from which  $a$  provably satisfies  $p$ . Hence  $a(n) = f(n)$  for every  $n$  in the claimed range.

**Theorem 1.** *The conjecture displayed above is true.*

### 3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly  $q$  says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.